

NYCU Computer Vision 2026 HW4

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1. Introduction

Image restoration is a fundamental low-level computer vision task that aims to recover high-quality clean images from their degraded counterparts. In real-world scenarios, images can be corrupted by various types of degradation such as rain streaks, snow, haze, and noise. Traditional single-degradation models require knowing the degradation type in advance and training separate models for each, which is impractical for real-world deployment. This work tackles the All-in-One image restoration problem for rain and snow degradation jointly. Specifically, given a degraded input image (either rain-corrupted or snow-corrupted), the model is required to produce a clean restored image without any prior knowledge of the degradation type. The training set consists of 1,600 rain and 1,600 snow degraded/clean image pairs, and the test set contains 100 images of unknown degradation type. We adopt PromptIR [1] as our backbone, a NeurIPS 2023 method that uses learned prompt vectors to dynamically encode degradation-specific information and guide the restoration network. We further introduce four modifications to improve restoration quality, described in detail in Section 2.

2. Method

2.1 PromptIR

Baseline PromptIR is built upon a U-Net-style hierarchical Transformer encoder-decoder. The encoder progressively downsamples the input feature maps through three scales using Pixel Unshuffle, while the decoder upsamples back to the original resolution via Pixel Shuffle. Each scale contains multiple Transformer blocks with two key components:

Multi-DConv Head Transposed Self-Attention (MDTA): Unlike standard spatial self-attention which has $O(N^2)$ complexity, MDTA applies attention across the channel dimension of transposed feature maps, making it computationally efficient for high-resolution inputs. Depth-wise convolutions are applied to Q, K, V projections to capture local context.

Gated Depth-wise Feed-Forward Network (GDFN): A gating mechanism controls information flow through the FFN, allowing the network to suppress less useful features and retain restoration-relevant activations.

2.2 Proposed Modifications

We introduce four modifications to the PromptIR baseline to improve performance on the rain/snow restoration task:

#	Modification	Details	Motivation
1	Channel Attention (SE Block)	Added after each FFN in every Transformer block. Uses avg+max pooling with reduction ratio 8.	Recalibrates channel-wise feature responses, helping the model focus on foreground textures.
2	FFT Frequency Loss	$L_{total} = L1 + 0.05 * L_{fft}$ where $L_{fft} = L1(FFT(pred) , FFT(target))$	Supervises high-frequency component recovery (edges, textures) beyond pixel-level L1 loss.
3	Longer Prompt (5 $\rightarrow$ 10)	Prompt length increased from 5 to 10 vectors per scale.	Richer encoding capacity to distinguish rain/snow degradation patterns.
4	Residual Scaling	Each block has a learnable scalar (init=0.1) on the FFN residual branch.	Stabilizes early training by initially suppressing FFN outputs.

2.3 Training Setup

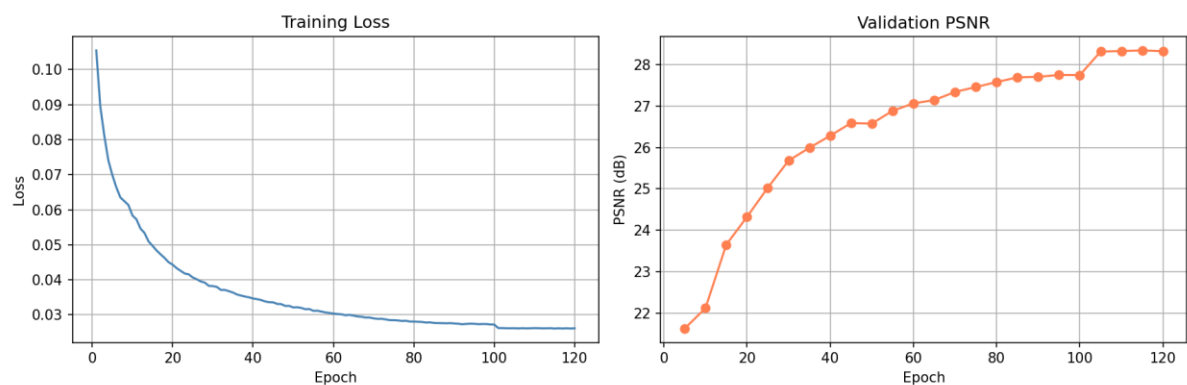
The model is trained from scratch for 100 epochs on the combined rain and snow training set (3,200 pairs total). We use the AdamW optimizer with an initial learning rate of 2×10^{-4} , weight decay 10^{-4} , and a cosine annealing schedule with 10 warm-up epochs. Input patches of size 128×128 are randomly cropped from training images. Data augmentation includes random horizontal/vertical flips and 90-degree rotations. Mixed-precision training (fp16) is applied for efficiency on an NVIDIA T4 GPU via Google Colab.

Hyperparameter	Value
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.999$, $wd=1e-4$)
Learning Rate	2×10^{-4} with cosine annealing

Warm-up Epochs	10
Total Epochs	120 (100 @ patch 128, then 20 @ patch 192)
Batch Size	4 (patch 128) / 2 (patch 192)
Patch Size	128×128
Loss	L1 + 0.05 × FFT Frequency Loss
Model Params	~60M

3. Results

(1) Training Curve:



Above shows the training loss and validation PSNR over 120 epochs. The loss decreases smoothly from 0.10 to 0.027 without oscillation. The validation PSNR rises steadily from 21.7 dB to 28.3 dB with no sign of overfitting, indicating that continued training would likely yield further improvement.

(2) Leaderboard Score: 29.68dB

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Submission	Epoch	Strategy	PSNR (dB)
Baseline	10	Standard inference	26.14
Continued training	50	Standard inference	28.54
Continued training	100	Standard inference	29.31
+ Patch 192 fine-tune	120	Standard inference	29.40

+ 8-fold D4 TTA	120	TTA + np rint rounding	29.68
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4. Additional Experiments

4.1 Fine-tuning with Larger Patch Size (128 $\rightarrow$ 192)

Hypothesis.

Training with small 128 $\times$ 128 patches may limit the model's ability to capture long-range rain streak patterns and large snow clusters, since their spatial extent often exceeds the crop window. We hypothesize that continuing training on larger 192 $\times$ 192 patches after initial convergence at 128 $\times$ 128 will expose the model to broader spatial context and improve restoration of large-scale degradation structures.

Setup.

After training to epoch 100 with patch size 128 (batch size 4), we resume training for an additional 20 epochs with patch size 192 (batch size 2, to maintain similar GPU memory usage). All other hyperparameters remain unchanged.

Results.

The CodaBench PSNR improved from 29.31 dB (epoch 100, patch 128) to 29.40 dB (epoch 120, patch 192), a gain of +0.09 dB.

Implication.

Larger patch fine-tuning provides a modest but consistent improvement. The gain likely comes from the model learning to handle spatially extended degradation patterns that are underrepresented in small crops. This is consistent with prior work showing that progressive patch size increase during training benefits image restoration models [2].

4.2 8-fold D4 Test-Time Augmentation (TTA)

Hypothesis.

Image restoration models may produce slightly different outputs depending on the orientation of the input, since convolutional and attention operations are not perfectly equivariant to rotation and reflection. By averaging predictions over multiple

augmented versions of the same input, we can reduce this variance and obtain a more stable, higher-quality restoration.

Setup.

At inference time, we apply 8 geometric transformations from the D4 symmetry group: 4 rotations ($0^\circ, 90^\circ, 180^\circ, 270^\circ$) $\times$ 2 reflections (original, horizontal flip). For each test image, the model produces 8 predictions; each is inverse-transformed back to the original orientation, and the 8 outputs are averaged pixel-wise. We also replace plain floor truncation with `np rint` (round-half-up) when converting float predictions to `uint8`, eliminating the systematic -0.5 LSB bias.

Results.

TTA improved PSNR from 29.40 dB to 29.68 dB, a gain of +0.28 dB, at no additional training cost.

Implication.

TTA provides a free +0.28 dB improvement without any retraining. This is a practically significant gain given the score distribution on the leaderboard. The result confirms that the model's outputs are not perfectly orientation-invariant, and that ensemble averaging across the D4 group effectively compensates for this. Combining both experiments (patch fine-tuning + TTA) yields a total improvement of +0.37 dB over the epoch-100 baseline.

5. References

- [1] Potlapalli, V., Zamir, S. W., Khan, S., & Khan, F. (2023). PromptIR: Prompting for All-in-One Blind Image Restoration. *Advances in Neural Information Processing Systems (NeurIPS)*, 36.
- [2] Zamir, S. W., Arora, A., Khan, S., Hayat, M., Khan, F. S., & Yang, M. H. (2022). Restormer: Efficient Transformer for High-Resolution Image Restoration. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- [3] Hu, J., Shen, L., & Sun, G. (2018). Squeeze-and-Excitation Networks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*

(CVPR).

[4] Jiang, Y., Chang, S., & Wang, Z. (2021). TransGAN: Two Pure Transformers Can Make One Strong GAN, and That Can Scale Up. NeurIPS.

[5] Focal Frequency Loss for Image Reconstruction and Synthesis. (2021).
Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV).

6. Code Reliability

```
PS C:\Users\user\Desktop\DL\HW4> python -m flake8 losses.py
PS C:\Users\user\Desktop\DL\HW4> python -m flake8 metrics.py
PS C:\Users\user\Desktop\DL\HW4> python -m flake8 model.py
PS C:\Users\user\Desktop\DL\HW4> python -m flake8 dataset.py
PS C:\Users\user\Desktop\DL\HW4> python -m flake8 train.py
PS C:\Users\user\Desktop\DL\HW4> python -m flake8 infer.py
PS C:\Users\user\Desktop\DL\HW4> 
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GitHub: <https://github.com/swimming-in-cs/NYCU-Computer-Vision-2026-HW4>